

Module 2 R Practice

Ruchika Patidar

2024-03-08

Introduction:

Age determination in abalone, involving microscopic examination of shell rings, is time-consuming. This report delves into predicting abalone age through accessible physical measurements, utilizing a dataset of 4176 records, encompassing shell dimensions, body weight, length, and ring count. The objective is to formulate predictive models, and the report provides a descriptive analysis of each variable.

```
abalone <- read.csv("abalone.data")
```

RStudio: Notebook Output

Description: df [4,176 x 9]

M <chr>	X0.455 <dbl>	X0.365 <dbl>	X0.095 <dbl>	X0.514 <dbl>	X0.2245 <dbl>	X0.101 <dbl>	X0.15 <dbl>	X15 <int>
M	0.350	0.265	0.090	0.2255	0.0995	0.0485	0.0700	7
F	0.530	0.420	0.135	0.6770	0.2565	0.1415	0.2100	9
M	0.440	0.365	0.125	0.5160	0.2155	0.1140	0.1550	10
I	0.330	0.255	0.080	0.2050	0.0895	0.0395	0.0550	7
I	0.425	0.300	0.095	0.3515	0.1410	0.0775	0.1200	8
F	0.530	0.415	0.150	0.7775	0.2370	0.1415	0.3300	20
F	0.545	0.425	0.125	0.7680	0.2940	0.1495	0.2600	16
M	0.475	0.370	0.125	0.5095	0.2165	0.1125	0.1650	9
F	0.550	0.440	0.150	0.8945	0.3145	0.1510	0.3200	19
F	0.525	0.380	0.140	0.6065	0.1940	0.1475	0.2100	14
M	0.430	0.350	0.110	0.4060	0.1675	0.0810	0.1350	10
M	0.490	0.380	0.135	0.5415	0.2175	0.0950	0.1900	11
F	0.535	0.405	0.145	0.6845	0.2725	0.1710	0.2050	10
F	0.470	0.355	0.100	0.4755	0.1675	0.0805	0.1850	10
M	0.500	0.400	0.130	0.6645	0.2580	0.1330	0.2400	12
I	0.355	0.280	0.085	0.2905	0.0950	0.0395	0.1150	7
F	0.440	0.340	0.100	0.4510	0.1880	0.0870	0.1300	10
M	0.365	0.295	0.080	0.2555	0.0970	0.0430	0.1000	7
M	0.450	0.320	0.100	0.3810	0.1705	0.0750	0.1150	9
M	0.355	0.280	0.095	0.2455	0.0955	0.0620	0.0750	11
I	0.380	0.275	0.100	0.2255	0.0800	0.0490	0.0850	10
F	0.565	0.440	0.155	0.9395	0.4275	0.2140	0.2700	12

1–22 of 4,176 rows

Previous123456...46Next

Renaming Columns

```
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

abalone <- abalone %>%
  rename(Sex = M,
         Length = X0.455,
         Diameter = X0.365,
         Height = X0.095,
         Whole_weight = X0.514,
         Shucked_weight = X0.2245,
         Viscera_weight = X0.101,
         Shell_weight = X0.15,
         Rings = X15)
```

RStudio: Notebook Output

Description: df [4,176 x 9]

Sex <chr>	Length <dbl>	Diameter <dbl>	Height <dbl>	Whole_weight <dbl>	Shucked_weight <dbl>	Viscera_weight <dbl>	Shell_weight <dbl>	Rings <int>
M	0.350	0.265	0.090	0.2255	0.0995	0.0485	0.0700	7
F	0.530	0.420	0.135	0.6770	0.2565	0.1415	0.2100	9
M	0.440	0.365	0.125	0.5160	0.2155	0.1140	0.1550	10
I	0.330	0.255	0.080	0.2050	0.0895	0.0395	0.0550	7
I	0.425	0.300	0.095	0.3515	0.1410	0.0775	0.1200	8
F	0.530	0.415	0.150	0.7775	0.2370	0.1415	0.3300	20
F	0.545	0.425	0.125	0.7680	0.2940	0.1495	0.2600	16
M	0.475	0.370	0.125	0.5095	0.2165	0.1125	0.1650	9
F	0.550	0.440	0.150	0.8945	0.3145	0.1510	0.3200	19
F	0.525	0.380	0.140	0.6065	0.1940	0.1475	0.2100	14
M	0.430	0.350	0.110	0.4060	0.1675	0.0810	0.1350	10
M	0.490	0.380	0.135	0.5415	0.2175	0.0950	0.1900	11
F	0.535	0.405	0.145	0.6845	0.2725	0.1710	0.2050	10
F	0.470	0.355	0.100	0.4755	0.1675	0.0805	0.1850	10
M	0.500	0.400	0.130	0.6645	0.2580	0.1330	0.2400	12
I	0.355	0.280	0.085	0.2905	0.0950	0.0395	0.1150	7
F	0.440	0.340	0.100	0.4510	0.1880	0.0870	0.1300	10
M	0.365	0.295	0.080	0.2555	0.0970	0.0430	0.1000	7
M	0.450	0.320	0.100	0.3810	0.1705	0.0750	0.1150	9
M	0.355	0.280	0.095	0.2455	0.0955	0.0620	0.0750	11
I	0.380	0.275	0.100	0.2255	0.0800	0.0490	0.0850	10
F	0.565	0.440	0.155	0.9395	0.4275	0.2140	0.2700	12

1-22 of 4,176 rows

Previous **1** 2 3 4 5 6 ... 46 Next

Length table Of Abalone

```
length_summary <- as.data.frame(table(abalone$Length))
colnames(length_summary) <- c("Length", "Frequency")
```

Length <fctr>	Frequency <int>
0.075	1
0.11	1
0.13	2
0.135	1
0.14	2
0.15	1
0.155	3
0.16	4
0.165	5
0.17	3
0.175	5
0.18	4
0.185	4
0.19	4
0.195	3
0.2	6
0.205	5
0.21	6
0.215	5
0.22	2
0.225	4
0.23	4

1-22 of 134 rows

Sex data table

```
sex_counts <- table(abalone$Sex)
sex_counts

##
##      F      I      M
## 1307 1342 1527

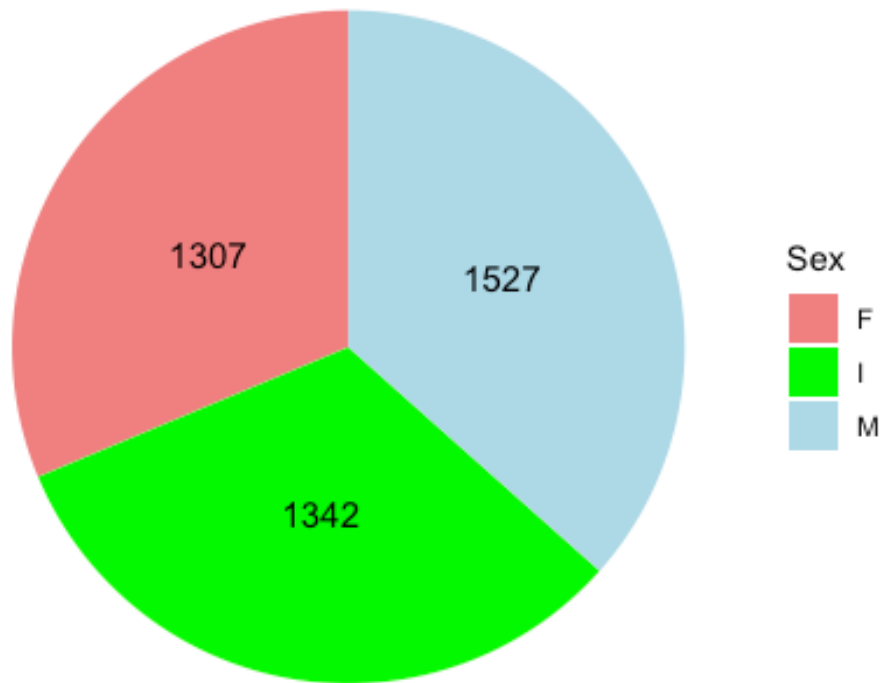
sex_data <- data.frame(Sex = names(sex_counts), Count =
as.numeric(sex_counts))
sex_data

##   Sex Count
## 1   F  1307
## 2   I  1342
## 3   M  1527
```

Sex Distribution Pie chart

```
library(ggplot2)
sex_colors <- c("F" = "lightcoral", "M" = "lightblue", "I" = "green")
pie_chart <- ggplot(sex_data, aes(x = "", y = Count, fill = Sex)) +
  geom_bar(stat = "identity", width = 1) +
  geom_text(aes(label = Count), position = position_stack(vjust = 0.5), color
= "black", size = 4) + # Add count labels
  coord_polar("y", start = 0) +
  labs(title = "Distribution of Sex in Abalone Dataset",
       fill = "Sex",
       x = NULL,
       y = NULL) +
  scale_fill_manual(values = sex_colors) +
  theme_void()
pie_chart
```

Distribution of Sex in Abalone Dataset



Descriptive Statistics for the Diameter Variable

```
library(psych)

##
## Attaching package: 'psych'

## The following objects are masked from 'package:ggplot2':
##
##      %+%, alpha

describe(abalone$Diameter)

##      vars      n mean  sd median trimmed mad  min  max range  skew kurtosis se
## X1       1 4176 0.41 0.1   0.42   0.41 0.1 0.06 0.65   0.6 -0.61   -0.05  0

diameter_descriptive <- data.frame(
  Statistics = c("Number of Observations (n)", "Mean", "Standard Deviation
(SD)",
                "Median", "Trimmed Mean", "Median Absolute Deviation (MAD)",
                "Minimum", "Maximum", "Range", "Skewness", "Kurtosis",
"Standard Error (SE)"),
  Values = c(4176, 0.41, 0.1, 0.42, 0.41, 0.1, 0.06, 0.65, 0.6, -0.61, -0.05,
0)
)
knitr::kable(diameter_descriptive, align = "l", caption = "Descriptive
Statistics for the 'Diameter' Variable")
```

Descriptive Statistics for the 'Diameter' Variable

Statistics	Values
Number of Observations (n)	4176.00
Mean	0.41
Standard Deviation (SD)	0.10
Median	0.42
Trimmed Mean	0.41
Median Absolute Deviation (MAD)	0.10
Minimum	0.06
Maximum	0.65
Range	0.60
Skewness	-0.61
Kurtosis	-0.05
Standard Error (SE).	0.00

Descriptive Statistics for the Height Variable

```
describe(abalone$Height)

##      vars      n mean   sd median trimmed  mad min  max range skew kurtosis se
## X1       1 4176 0.14 0.04   0.14   0.14 0.04   0 1.13  1.13 3.13    75.92  0

height_descriptive <- data.frame(
  Statistics = c("Number of Observations (n)", "Mean", "Standard Deviation
(SD)",
                "Median", "Trimmed Mean", "Median Absolute Deviation (MAD)",
                "Minimum", "Maximum", "Range", "Skewness", "Kurtosis",
"Standard Error (SE)"),
  Values = c(4176, 0.14, 0.04, 0.14, 0.14, 0.04, 0, 1.13, 1.13, 3.13, 75.92,
0)
)
knitr::kable(height_descriptive, align = "l", caption = "Descriptive
Statistics for the 'Height' Variable")
```

Descriptive Statistics for the 'Height' Variable

Statistics	Values
Number of Observations (n)	4176.00
Mean	0.14
Standard Deviation (SD)	0.04
Median	0.14
Trimmed Mean	0.14
Median Absolute Deviation (MAD)	0.04
Minimum	0.00
Maximum	1.13
Range	1.13
Skewness	3.13
Kurtosis	75.92
Standard Error (SE)	0.00

Descriptive Statistics for Whole_weight Variable

```
describe(abalone$Whole_weight)

##      vars      n mean   sd median trimmed  mad min  max range skew kurtosis
se
## X1      1 4176 0.83 0.49    0.8    0.8 0.53   0 2.83  2.82 0.53   -0.03
0.01

whole_weight_descriptive <- data.frame(
  Statistics = c("Number of Observations (n)", "Mean", "Standard Deviation
(SD)",
                "Median", "Trimmed Mean", "Median Absolute Deviation (MAD)",
                "Minimum", "Maximum", "Range", "Skewness", "Kurtosis",
"Standard Error (SE)"),
  Values = c(4176, 0.83, 0.49, 0.8, 0.8, 0.53, 0, 2.83, 2.82, 0.53, -0.03,
0.01)
)
knitr::kable(whole_weight_descriptive, align = c("l", "c"), caption =
"Descriptive Statistics for 'Whole_weight' Variable")
```

Descriptive Statistics for 'Whole_weight' Variable

Statistics	Values
Number of Observations (n)	4176.00
Mean	0.83
Standard Deviation (SD)	0.49
Median	0.80
Trimmed Mean	0.80
Median Absolute Deviation (MAD)	0.53
Minimum	0.00
Maximum	2.83
Range	2.82
Skewness	0.53
Kurtosis	-0.03
Standard Error (SE)	0.01

Descriptive Statistics for Shucked_weight Variable

```
describe(abalone$Shucked_weight)

##      vars      n mean   sd median trimmed  mad min  max range skew kurtosis se
## X1       1 4176 0.36 0.22   0.34   0.34 0.23   0 1.49  1.49 0.72    0.59  0

shucked_weight_descriptive <- data.frame(
  Statistics = c("Number of Observations (n)", "Mean", "Standard Deviation
(SD)",
                "Median", "Trimmed Mean", "Median Absolute Deviation (MAD)",
                "Minimum", "Maximum", "Range", "Skewness", "Kurtosis",
"Standard Error (SE)"),
  Values = c(4176, 0.36, 0.22, 0.34, 0.34, 0.23, 0, 1.49, 1.49, 0.72, 0.59,
0)
)
knitr::kable(shucked_weight_descriptive, align = c("l", "c"), caption =
"Descriptive Statistics for 'Shucked_weight' Variable")
```

Descriptive Statistics for 'Shucked_weight' Variable

Statistics	Values
Number of Observations (n)	4176.00
Mean	0.36
Standard Deviation (SD)	0.22
Median	0.34
Trimmed Mean	0.34
Median Absolute Deviation (MAD)	0.23
Minimum	0.00
Maximum	1.49
Range	1.49
Skewness	0.72
Kurtosis	0.59
Standard Error (SE)	0.00

Descriptive Statistics for Viscera_weight Variable

```
describe(abalone$Viscera_weight)

##      vars      n mean   sd median trimmed  mad min  max range skew kurtosis se
## X1       1 4176 0.18 0.11   0.17   0.17 0.12   0 0.76  0.76 0.59    0.08  0

viscera_weight_descriptive <- data.frame(
  Statistics = c("Number of Observations (n)", "Mean", "Standard Deviation
(SD)",
                "Median", "Trimmed Mean", "Median Absolute Deviation (MAD)",
                "Minimum", "Maximum", "Range", "Skewness", "Kurtosis",
"Standard Error (SE)"),
  Values = c(4176, 0.18, 0.11, 0.17, 0.17, 0.12, 0, 0.76, 0.76, 0.59, 0.08,
0)
)
knitr::kable(viscera_weight_descriptive, align = c("l", "c"), caption =
"Descriptive Statistics for 'Viscera_weight' Variable")
```

Descriptive Statistics for 'Viscera_weight' Variable

Statistics	Values
Number of Observations (n)	4176.00
Mean	0.18
Standard Deviation (SD)	0.11
Median	0.17
Trimmed Mean	0.17
Median Absolute Deviation (MAD)	0.12
Minimum	0.00
Maximum	0.76
Range	0.76
Skewness	0.59
Kurtosis	0.08
Standard Error (SE)	0.00

Descriptive Statistics for Shell_weight Variable

```
describe(abalone$Shell_weight)

##      vars      n mean   sd median trimmed  mad min max range skew kurtosis se
## X1       1 4176 0.24 0.14   0.23   0.23 0.15   0  1    1 0.62    0.53  0

shell_weight_descriptive <- data.frame(
  Statistics = c("Number of Observations (n)", "Mean", "Standard Deviation (SD)",
    "Median", "Trimmed Mean", "Median Absolute Deviation (MAD)",
    "Minimum", "Maximum", "Range", "Skewness", "Kurtosis",
    "Standard Error (SE)"),
  Values = c(4176, 0.24, 0.14, 0.23, 0.23, 0.15, 0, 1, 1, 0.62, 0.53, 0)
)
knitr::kable(shell_weight_descriptive, align = c("l", "c"), caption =
"Descriptive Statistics for 'Shell_weight' Variable")
```

Descriptive Statistics for 'Shell_weight' Variable

Statistics	Values
Number of Observations (n)	4176.00
Mean	0.24
Standard Deviation (SD)	0.14
Median	0.23
Trimmed Mean	0.23
Median Absolute Deviation (MAD)	0.15
Minimum	0.00
Maximum	1.00
Range	1.00
Skewness	0.62
Kurtosis	0.53
Standard Error (SE)	0.00

Descriptive Statistics for Rings Variable

```
describe(abalone$Rings)

##      vars      n mean   sd median trimmed  mad min max range skew kurtosis
se
## X1      1 4176 9.93 3.22      9   9.64 2.97   1  29   28 1.11      2.33
0.05

rings_descriptive <- data.frame(
  Statistics = c("Number of Observations (n)", "Mean", "Standard Deviation
(SD)",
                "Median", "Trimmed Mean", "Median Absolute Deviation (MAD)",
                "Minimum", "Maximum", "Range", "Skewness", "Kurtosis",
"Standard Error (SE)"),
  Values = c(4176, 9.93, 3.22, 9, 9.64, 2.97, 1, 29, 28, 1.11, 2.33, 0.05)
)
knitr::kable(rings_descriptive, align = c("l", "c"), caption = "Descriptive
Statistics for 'Rings' Variable")
```

Descriptive Statistics for 'Rings' Variable

Statistics	Values
Number of Observations (n)	4176.00
Mean	9.93
Standard Deviation (SD)	3.22
Median	9.00
Trimmed Mean	9.64
Median Absolute Deviation (MAD)	2.97
Minimum	1.00
Maximum	29.00
Range	28.00
Skewness	1.11
Kurtosis	2.33
Standard Error (SE)	0.05

Descriptive Statistic table of length

```
describe(abalone$Length)

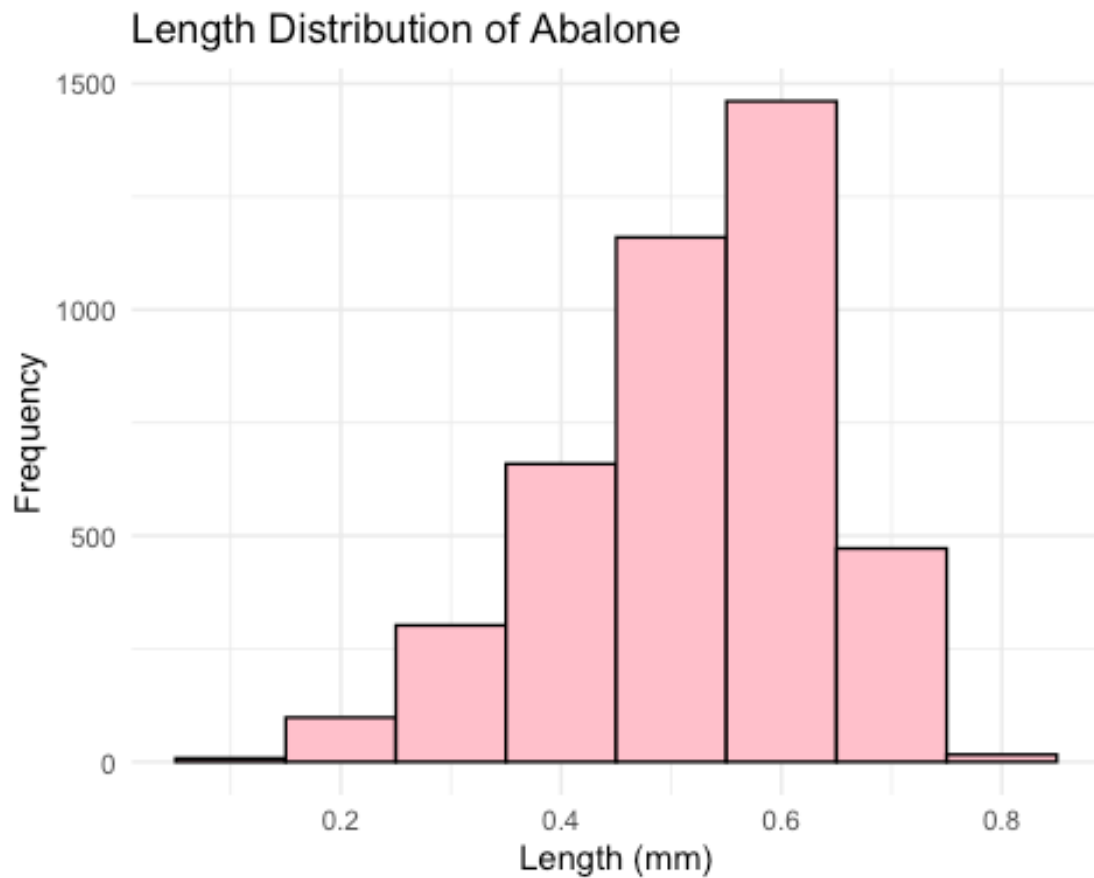
##      vars      n mean   sd median trimmed  mad   min   max range  skew kurtosis
## X1      1 4176 0.52 0.12   0.54   0.53 0.12 0.07 0.81  0.74 -0.64    0.06
##
length_descriptive <- data.frame(
  Statistics = c("Number of Observations (n)", "Mean", "Standard Deviation
(SD)",
                "Median", "Trimmed Mean", "Median Absolute Deviation (MAD)",
                "Minimum", "Maximum", "Range", "Skewness", "Kurtosis",
"Standard Error (SE)"),
  Values = c(4176, 0.52, 0.12, 0.54, 0.53, 0.12, 0.07, 0.81, 0.74, -0.64,
0.06, 0)
)
knitr::kable(length_descriptive, align = "l", caption = "Descriptive
Statistics for the 'Length' Variable")
```

Descriptive Statistics for the 'Length' Variable

Statistics	Values
Number of Observations (n)	4176.00
Mean	0.52
Standard Deviation (SD)	0.12
Median	0.54
Trimmed Mean	0.53
Median Absolute Deviation (MAD)	0.12
Minimum	0.07
Maximum	0.81
Range	0.74
Skewness	-0.64
Kurtosis	0.06
Standard Error (SE)	0.00

Distribution of Length

```
ggplot(abalone, aes(x = Length)) +  
  geom_histogram(binwidth = 0.1, fill = "pink", color = "black") +  
  labs(title = "Length Distribution of Abalone",  
        x = "Length (mm)",  
        y = "Frequency") +  
  theme_minimal()
```

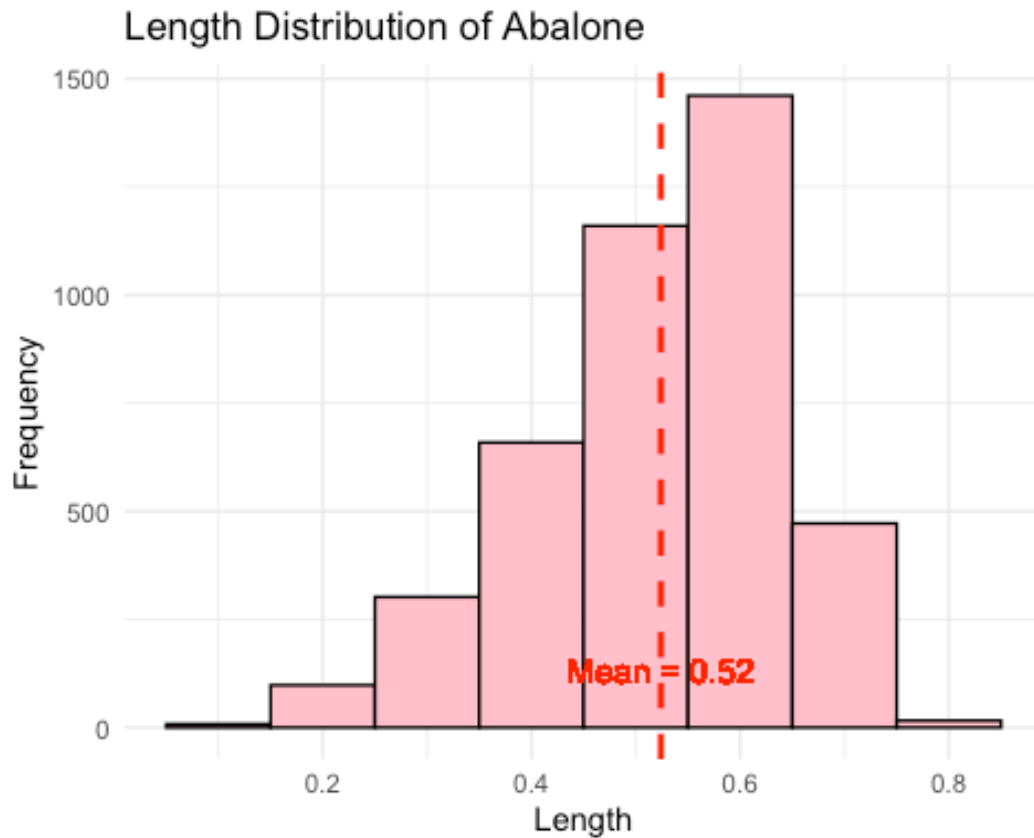


Mean Distribution length

```
mean_Length <- mean(abalone$Length, na.rm = TRUE)
print(mean_Length)

## [1] 0.5240086

ggplot(abalone, aes(x = Length)) +
  geom_histogram(binwidth = 0.1, fill = "pink", color = "black") +
  geom_vline(xintercept = mean_Length, color = "red", linetype = "dashed",
size = 1) +
  geom_text(aes(x = mean_Length, label = paste("Mean =", round(mean_Length,
2))),
            y = 50, vjust = -1, color = "red", size = 4) +
  labs(title = "Length Distribution of Abalone",
        x = "Length",
        y = "Frequency") +
  theme_minimal()
```

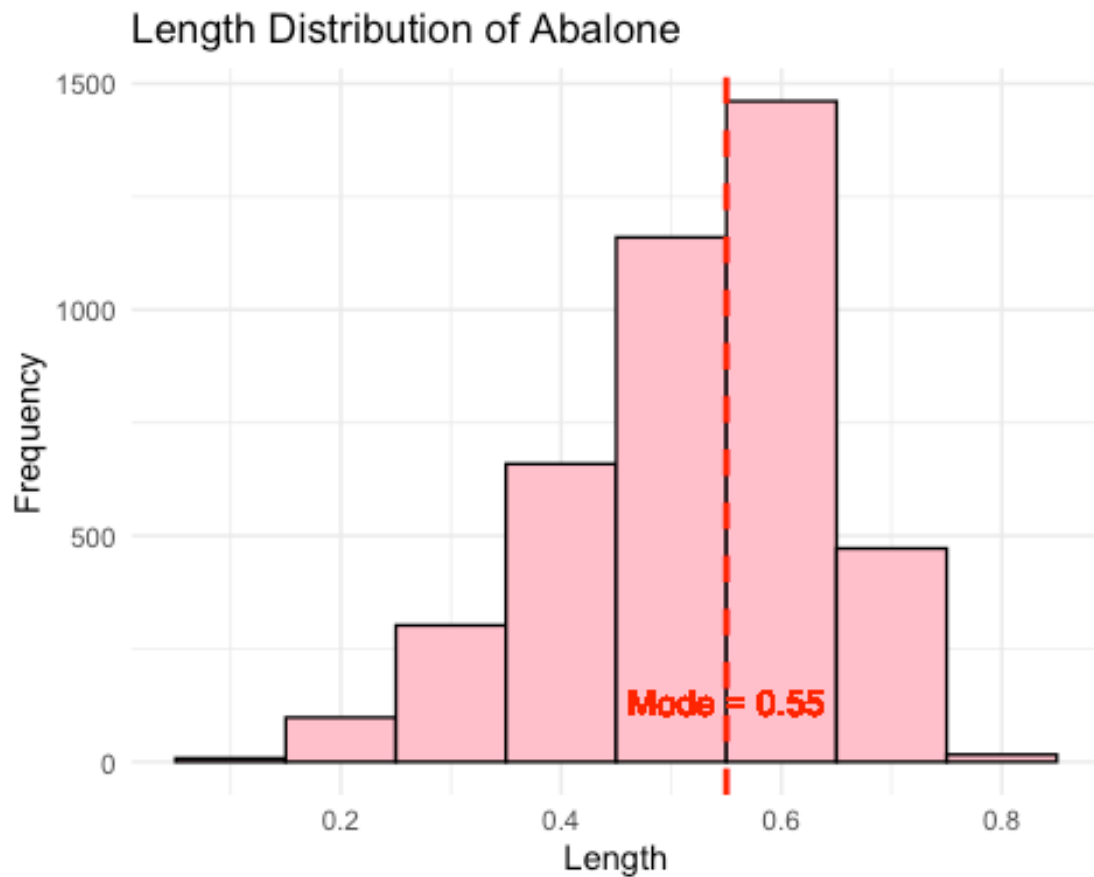


Mode Distribution Length

```
Mode <- function(x) {
  ux <- unique(x)
  ux[which.max(tabulate(match(x, ux)))]
}
mode_Length <- Mode (abalone$Length)
print(mode_Length)

## [1] 0.55

ggplot(abalone, aes(x = Length)) +
  geom_histogram(binwidth = 0.1, fill = "pink", color = "black") +
  geom_vline(xintercept = mode_Length, color = "red", linetype = "dashed",
size = 1) +
  geom_text(aes(x = mode_Length, label = paste("Mode =", mode_Length)),
    y = 50, vjust = -1, color = "red", size = 4) +
  labs(title = "Length Distribution of Abalone",
    x = "Length",
    y = "Frequency") +
  theme_minimal()
```

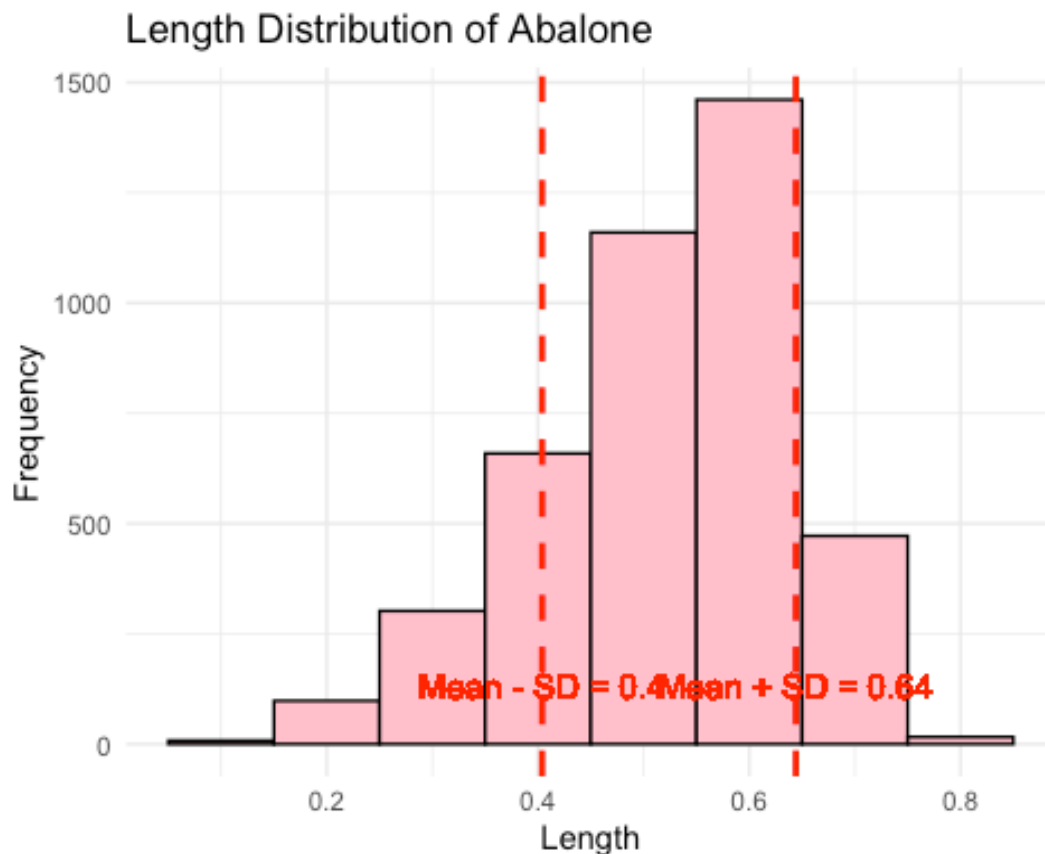


Standard deviation of length

```
sd_Length <- sd(abalone$Length)
print(sd_Length)

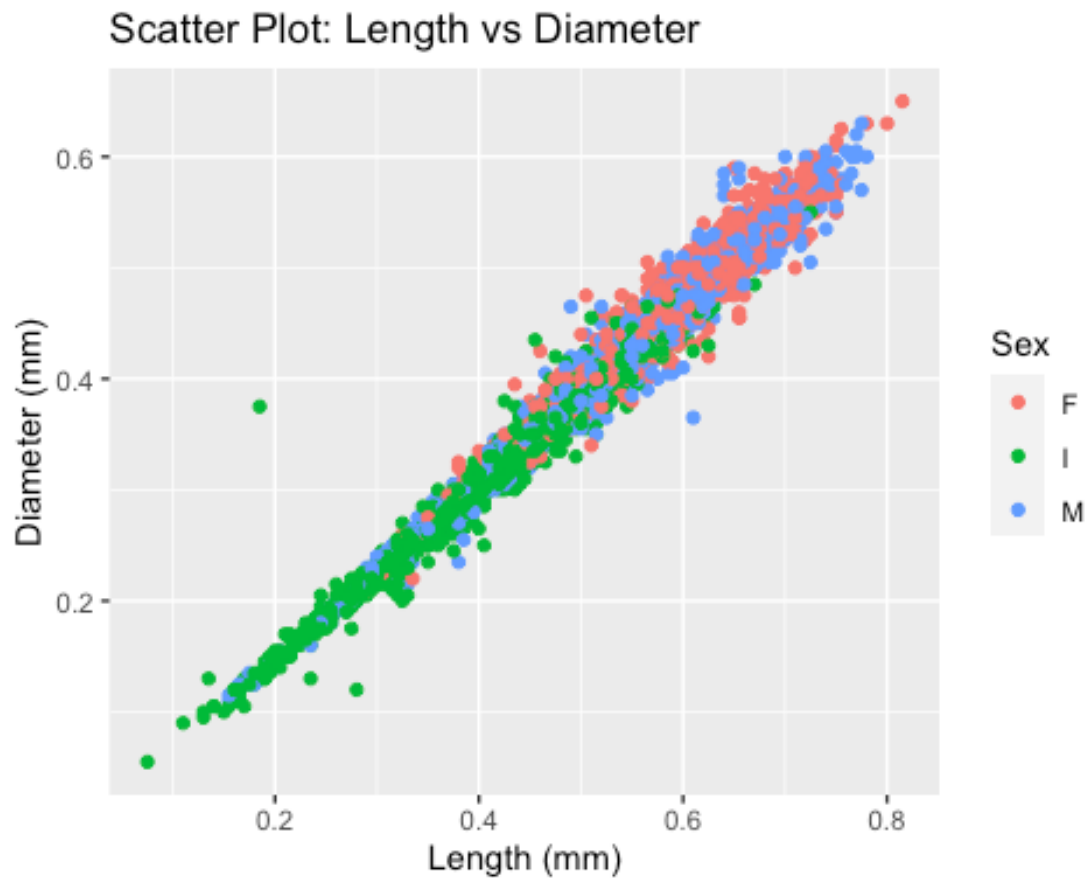
## [1] 0.1201025

ggplot(abalone, aes(x = Length)) +
  geom_histogram(binwidth = 0.1, fill = "pink", color = "black") +
  geom_vline(xintercept = mean(abalone$Length) + sd_Length, color = "red",
linetype = "dashed", size = 1) +
  geom_vline(xintercept = mean(abalone$Length) - sd_Length, color = "red",
linetype = "dashed", size = 1) +
  geom_text(aes(x = mean(abalone$Length) + sd_Length, label = paste("Mean +
SD =", round(mean(abalone$Length) + sd_Length, 2))),
y = 50, vjust = -1, color = "red", size = 4) +
  geom_text(aes(x = mean(abalone$Length) - sd_Length, label = paste("Mean -
SD =", round(mean(abalone$Length) - sd_Length, 2))),
y = 50, vjust = -1, color = "red", size = 4) +
  labs(title = "Length Distribution of Abalone",
x = "Length",
y = "Frequency") +
  theme_minimal()
```



Scatter Plot for Length vs Diameter

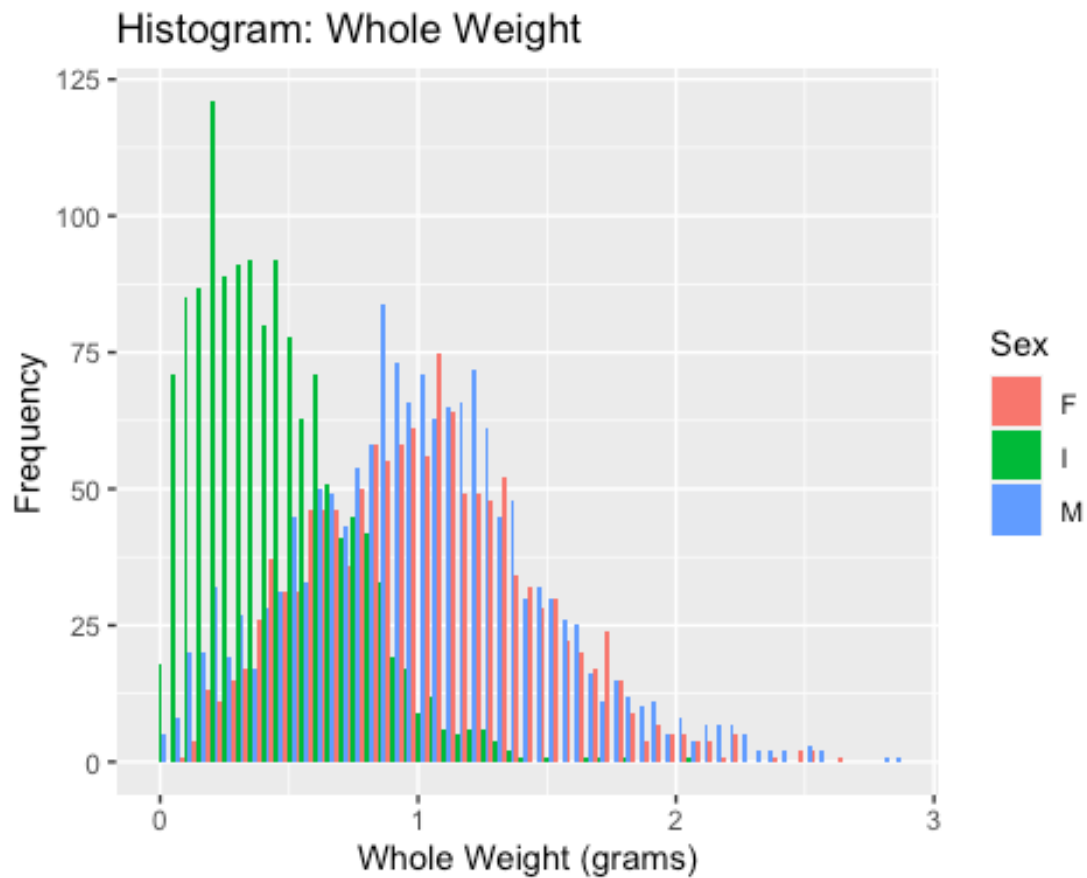
```
scatter_plot <- ggplot(abalone, aes(x = Length, y = Diameter, color = Sex)) +  
  geom_point() +  
  labs(title = "Scatter Plot: Length vs Diameter",  
        x = "Length (mm)",  
        y = "Diameter (mm)")  
scatter_plot
```



This scatter plot shows the relationship between the diameter and length of abalone, with different colors representing the sex of the abalone. Sex is a categorical variable, while diameter and length are continuous variables.

Histogram: Whole Weight

```
histogram_plot <- ggplot(abalone, aes(x = Whole_weight, fill = Sex)) +  
  geom_histogram(binwidth = 0.05, position = "dodge") +  
  labs(title = "Histogram: Whole Weight",  
       x = "Whole Weight (grams)",  
       y = "Frequency")  
histogram_plot
```

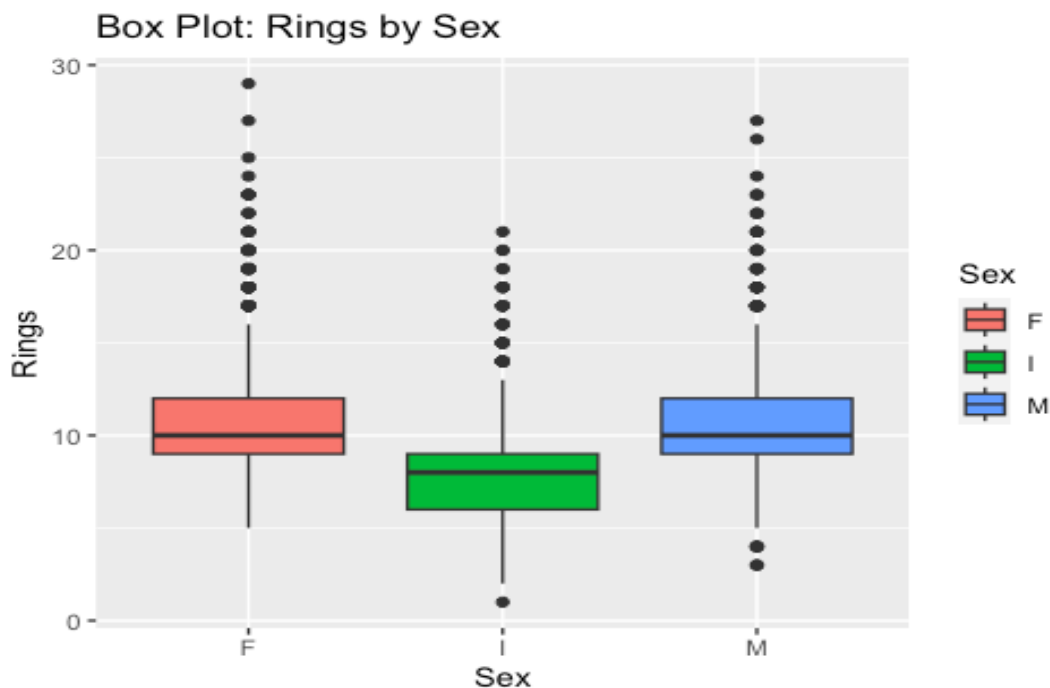


This graph suggests that the weight of infant abalone is generally lower compared to both male and female abalone.

How can you use boxplots to detect outliers?

Boxplots are visual snapshots of data that help us spot unusual values, called outliers. Outliers are those oddball data points that stand out from the rest and might mess up our analyses. A box in the middle of the plot – that's where most of the data hangs out. The lines (whiskers) around the box show the typical range of values. Boxplots can be used with any dataset that contains numerical variables, as they are effective for visualizing the distribution of numerical data and identifying outliers. In this example, a boxplot is used to compare the number of rings of abalone with the sex of the abalone. The boxplot displays the distribution of the number of rings for different sexes of abalone. Different colors are used to represent each sex: light coral for females, blue for males, and green for infants. The boxplot indicates that most abalone, regardless of sex, have a number of rings between 5 and 15.

```
box_plot <- ggplot(abalone, aes(x = Sex, y = Rings, fill = Sex)) +
  geom_boxplot() +
  labs(title = "Box Plot: Rings by Sex",
       x = "Sex",
       y = "Rings") +
  stat_summary(fun.data = "count", geom = "text", aes(label =
after_stat(count)), vjust = -0.5, size = 3, color = "black")
box_plot
```

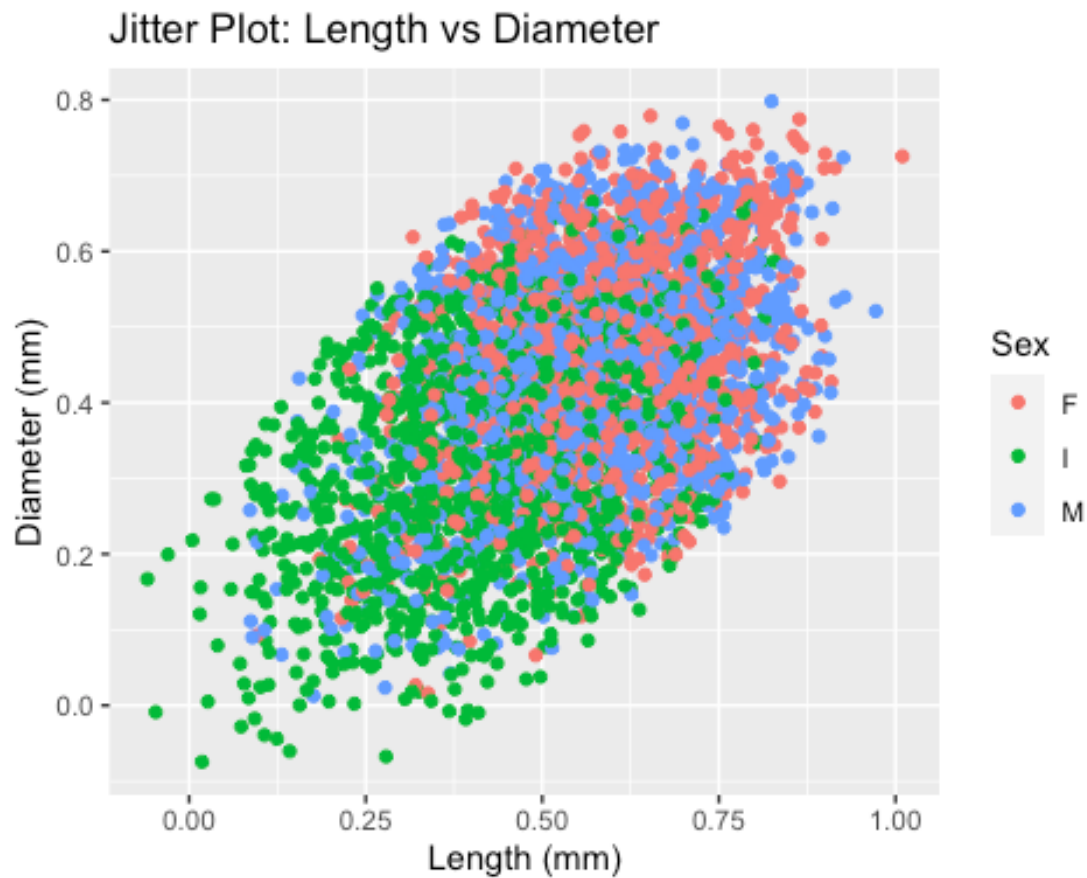


When do you use jitter chart?

Jitter plots are kind of scatter plot that are primarily used to avoid overlapping in situations where there are many data points with the same or similar values for the continuous variable within a category. Below are some example of jitter plots with Abalone dataset.

Jitter Plot for Length vs Diameter

```
jitter_plot_length_diameter <- ggplot(abalone, aes(x = Length, y = Diameter,  
color = Sex)) +  
  geom_jitter(width = 0.2, height = 0.2) +  
  labs(title = "Jitter Plot: Length vs Diameter",  
        x = "Length (mm)",  
        y = "Diameter (mm)")  
jitter_plot_length_diameter
```



Jitter Plot for Whole Weight vs Shucked Weight

```
jitter_plot_weight <- ggplot(abalone, aes(x = Whole_weight, y =  
Shucked_weight, color = Sex)) +  
geom_jitter(width = 0.2, height = 0.2) +  
  labs(title = "Jitter Plot: Whole Weight vs Shucked Weight",  
        x = "Whole Weight (grams)",  
        y = "Shucked Weight (grams)")  
jitter_plot_weight
```



Jitter Plot for Viscera Weight vs Shell Weight

```
jitter_plot_weight_viscera <- ggplot(abalone, aes(x = Viscera_weight, y =  
Shell_weight, color = Sex)) +  
  geom_jitter(width = 0.2, height = 0.2) +  
  labs(title = "Jitter Plot: Viscera Weight vs Shell Weight",  
        x = "Viscera Weight (grams)",  
        y = "Shell Weight (grams)")  
jitter_plot_weight_viscera
```

